

Day 43 - FTP & TFTP

Wednesday, September 9, 2026

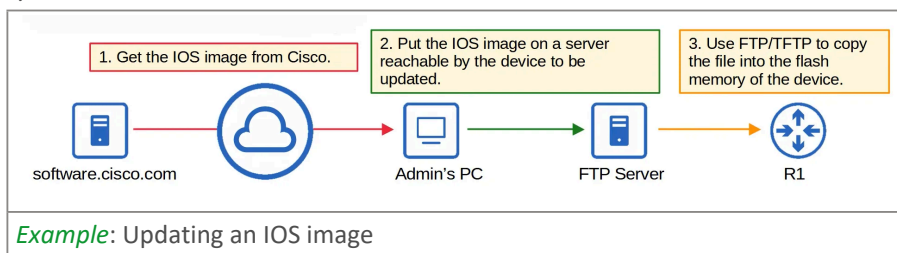
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FTP & TFTP Info

File Transfer Protocol (FTP) and Trivial File Transfer Protocol (TFTP) are *industry standard* protocols used to transfer files over a network

- **FTP** servers use port **20** for **FTP Data** connections and port **21** for **FTP Control** connections.
- **TFTP** servers use port **69**.
- They both use a client-server model.
 - Clients can use **FTP** or **TFTP** to copy files from a server.
 - Clients can use **FTP** or **TFTP** to copy files to a server.

For network engineers, the most common use for FTP/TFTP is to upgrade the operating system of a network device, as you can use it to download a newer version of IOS and then reboot the device with the new IOS image.



Trivial File Transfer Protocol (TFTP) General Information

TFTP is called '*Trivial*' because it is a simpler version of FTP and has only basic features.

- **TFTP** servers listen on port **69**
- Uses **UDP**, which is connectionless and doesn't provide reliability with retransmissions, however, **TFTP** has built-in features to ensure the delivery of files.
- Only allows a client to copy a file to or from a **TFTP** server.
- No authentication system (*Username/Password*), so servers will respond to all **TFTP** requests.
- No encryption, so all data is sent in plain text.
- Best used in controlled environments to quickly transfer small files.

TFTP Reliability

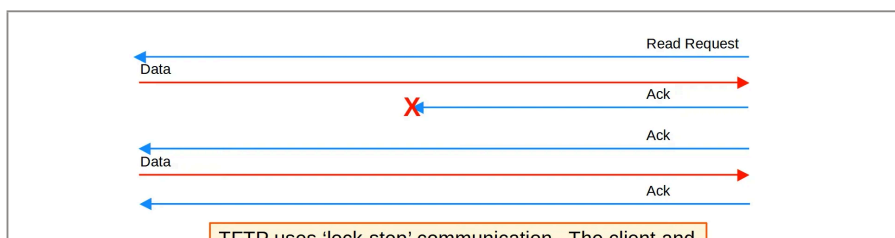
The TFTP protocol provides some reliability by having a built-in acknowledgement system for file transfers.

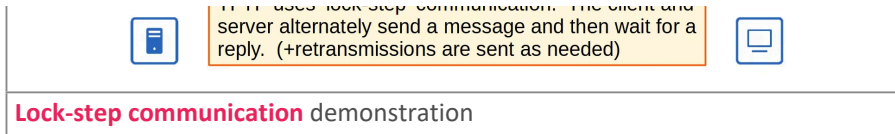
- Every **TFTP** data message is acknowledged with an **Ack** message:
 - If a client is transferring files to a server, the server will send **Ack** messages.
 - If the server is transferring files to a client, the client will send **Ack** messages.
- Timers are used, and if an expected message isn't received in time, the waiting device will re-send its previous message.

TFTP: Lock-step communication

This method of communicating in TFTP where one party sends the files and the other waits for an **Ack** reply is called Lock-step communication.

- The client and server alternately send a message and then wait for a reply.
- And retransmissions are sent as needed.





TFTP Connections

TFTP file transfers have three different phases:

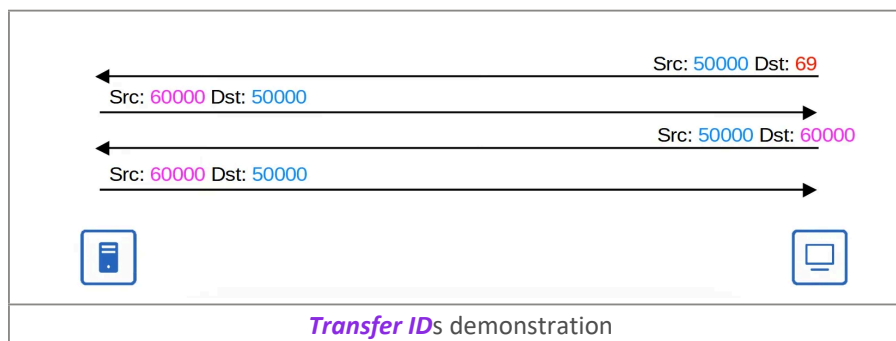
1. **Connection**
 - TFTP client sends a request and the server responds back, initializing the connection.
2. **Data Transfer**
 - The client and server exchange TFTP messages. One sends data while the other responds with **Ack** messages.
3. **Connection Termination**
 - After the last data message has been sent, a final **Ack** is sent to terminate the connection.



Extra: TFTP Transfer IDs

TFTP uses a **Transfer ID** system when sending files between the client and the server.

- When the client sends the first message to the server, the **source port** is a random ephemeral range port, and the **destination port** is **UDP 69**.
- This random port is called a '**Transfer Identifier (TID)**' and identifies the data transfer.
- The server then also selects a random **TID** to use as the **source port** when it replies, rather than port **69**.
- When the client sends the next message, it will change the **destination port** from port **69** to the newly established server **TID**.



File Transfer Protocol (FTP)

File Transfer Protocol (FTP) is the more advanced file sharing protocol.

- FTP servers use port **20** for **FTP Data** connections and port **21** for **FTP Control** connections.
- Uses **TCP**, which is connection-based and ensures the delivery of files
- Allows for file transfers, navigation of file directories, adding and removing directories, listing files, and etc.
- Has an authentication system (**Username/Password**).
- No encryption, so all data is sent in plain text.
- The client sends **FTP** commands to the server to perform these functions.

FTP Connections Types

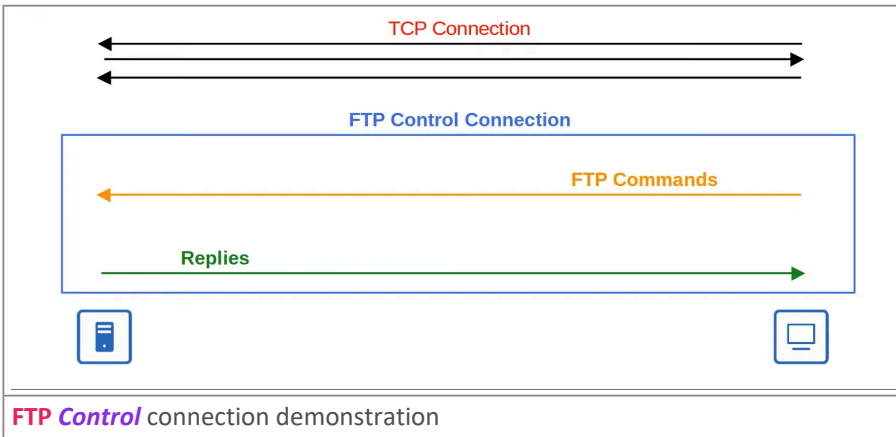
FTP uses two types of connections:

FTP uses two types of connections.

1. **FTP Control (TCP 21)**
 - Established and used to send **FTP** commands and replies.
2. **FTP Data (TCP 20)**
 - Established and terminated as needed when files or data are transferred between devices.

FTP Control Connections

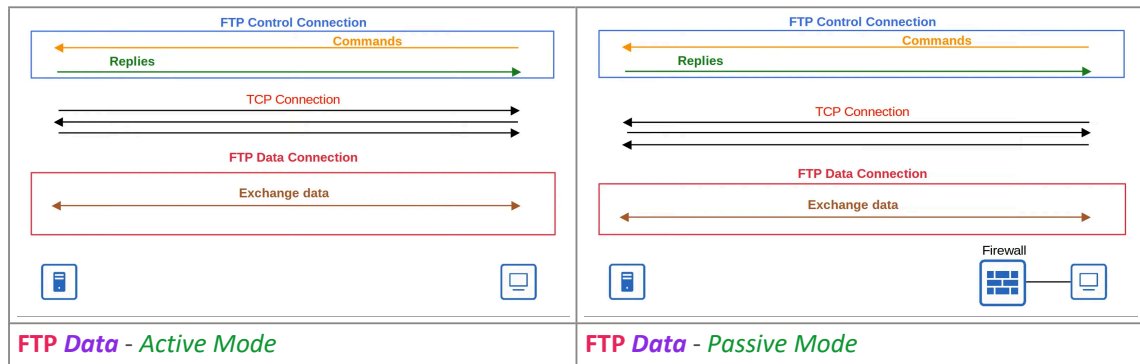
FTP **Control** is established between devices and used to send FTP orders (commands) from the client to the server.



FTP Data Connections

FTP **Data** connections are established and terminated as needed in order to send and receive files between the two participating parties.

- There are two types of **FTP Data** connection modes:
 1. **Active Mode**
 - In this mode, the server initiates the **TCP** connection.
 - This is the default method that **FTP Data** connections use.
 2. **Passive Mode**
 - In this mode, the client initiates the **TCP** connection.
 - It is often used when the client is behind a **firewall**, which could block incoming connections from the server.



Firewalls and connections

- Firewalls** usually don't permit 'outside' devices to initiate connections.
- In this case, **FTP passive mode** is used and the client (which is behind the **firewall**) initiates the **TCP** connection.

FTP VS. TFTP

FTP	TFTP
Uses TCP (ports 20 , 21)	Uses UDP (port 69)
Clients can use FTP commands to perform various actions (Copy files, add and remove directories, list directories, etc.)	Clients can only copy files to and from the server.

<i>Username/Password</i> authentication	No <i>Username/Password</i> authentication
Complex	Simple

General Cisco IOS Information

IOS File Systems

A file system is a way of controlling how data is stored and retrieved.

- You can see the file systems of a Cisco IOS device with the *show file systems* command.
- The most important file systems for the CCNA are:
 - Disk**
 - Storage devices such as *flash memory*.
 - Opaque**
 - Used for logical internal functions.
 - NVRAM**
 - Internal Non-volatile RAM. (which stays after the device has been shutdown)
 - The *startup-config* file is stored in here.
 - Network**
 - Represents external file systems.
 - For example, external **FTP/TFTP** servers.

Cisco IOS Updating (Configurations)

- show version*
 - Views the current version of IOS.
- show flash*
 - Shows the *flash memory*'s contents.
 - You can also view the identify IOS images by looking for the *.bin* file extension in the output.
- boot system FILEPATH*
 - Allows you to change the boot system IOS image.
 - Used for changing IOS versions.
 - (If no *file-path* is specified, the system will default to the first IOS image it finds)
- delete FILEPATH*
 - Deletes the specified file.
- copy SOURCE DESTINATION*
 - The general command used to copy files from source to destination
- copy tftp: flash:*
 - Used to copy files from a **TFTP** server to the *flash memory* of the device.
 - It asks for the server's *IP*, *source filename* and the *destination filename*.
- copy ftp: flash:*
 - Used to copy files from a **FTP** server to the *flash memory* of the device.
 - It asks for the server's *IP*, *source filename* and the *destination filename*.
 - If authentication is enabled, then you have to configure it through the *ip ftp username USERNAME* and *ip ftp password PASSWORD* commands.
- ip ftp username USERNAME*
 - Configures the **FTP username** used in **FTP** connections.
- ip ftp password PASSWORD*
 - Configures the **FTP password** used in **FTP** connections.